

Priyansh Paliwal

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Education

VIT Bhopal University

Bachelor of Technology (B.Tech)

Bhopal, India

Sept 2023 – June 2027

- CGPA: 9.02

Technical Skills

Programming Languages: Java, Python, C++, JavaScript, TypeScript

Frontend: HTML, CSS, React, Next.js, Tailwind CSS

Backend: RESTful APIs, FastAPI, Node.js, Prisma

AI & Machine Learning: RAG, Gemini API, Vector Search, ONNX

Database: PostgreSQL, SQL, ChromaDB, SQLite

Core Concepts: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Computer Networks, Operating Systems, System Design

Tools: Git, GitHub, Apache Kafka, Tree-Sitter, VS Code, Jupyter Notebook, Power BI

Projects

Real Estate Web App

Next.js, React, TypeScript, Tailwind CSS, PostgreSQL, Prisma

[Demo](#) | [GitHub](#)

- Built a full-stack real estate platform supporting 3 property categories with structured listings for pricing, dimensions, location, and property details.
- Developed 4 RESTful CRUD operations and integrated them with the frontend to enable end-to-end property listing management.
- Designed a normalized PostgreSQL database with Prisma ORM, modeling property data across 5+ key attributes including pricing, dimensions, coordinates, and listing metadata.
- Integrated Google Business Profile with the platform, enabling customer inquiries and issue reporting while improving local business discoverability.

Intelligent Codebase RAG Engine

Python, FastAPI, Next.js, Tree-sitter, ChromaDB, Gemini

[Demo](#) | [GitHub](#)

- Architected a Hybrid GraphRAG engine for GitHub repositories, eliminating broken syntax scopes by parsing Tree-sitter ASTs to extract complete function boundaries and 480+ call/import dependencies into SQLite.
- Developed a zero-cost local embedding pipeline using CPU-accelerated ONNX (all-MiniLM-L6-v2) inside ChromaDB, cutting repository ingestion time to under 4 seconds with automated noise filtering.
- Reduced end-to-end query latency by 20x (from 19s to 0.9s) by benchmarking inference models and deploying Google Gemini 3.5 Flash-Lite with dual-mode zero-hallucination guardrails.
- Built an asynchronous FastAPI REST API that retrieves relevant codebase context and delivers grounded, cited explanations to a Next.js frontend in under 1 second.

Real-Time E-commerce Analytics Pipeline

Python, Apache Kafka, SQL, Power BI

[Demo](#) | [GitHub](#)

- Addressed a 2-hour latency gap in sales data reporting that delayed inventory updates.
- Architected a real-time streaming pipeline using Apache Kafka to ingest live transactions into a SQL data warehouse.
- Implemented data transformation and validation workflows to ensure accurate and consistent transaction data for downstream analytics.
- Delivered real-time dashboards with under 5-second latency, driving a 15% increase in inventory efficiency.

Certifications

- Top 1% (Elite + Gold, 99%), Marketing Analytics – **NPTEL** Jan 2026 – Apr 2026
- Elite Certification, Introduction to Machine Learning – **NPTEL** Jan 2025 – Apr 2025
- Advanced Software Engineering and Dev Internship – **MPOnline Limited** May 2026 – Jul 2026
- AI/ML Internship – **MPOnline Limited** May 2026 – Jul 2026
- Bits and Bytes of Computer Networking – **Coursera**